THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A fair die is rolled once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{4}$

2. A fair die is rolled once. The probability of getting an odd prime number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

3. A bag contains 3 red, 9 blue and 3 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{1}{2}$

4. A bag contains 9 red, 3 blue and 3 green balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{2}{5}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{4}{5}$ **(d)** $\frac{3}{5}$

5. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking the letter A is: 1

(a) $\frac{2}{11}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{9}{11}$

6. For the sequence $t_n = n^2 - 2n + 3$, the value of t_7 is: 1

(a) 39 **(b)** 37 **(c)** 38 **(d)** 76

7. Which term of the sequence $t_n = 2n - 1$ is 137? 1

(a) $n = 68$ **(b)** $n = 69$ **(c)** $n = 70$ **(d)** $n = 71$

8. The 10th term of the AP $3, 8, 13, \ldots$ is: 1

(a) 48 **(b)** 96 **(c)** 49 **(d)** 47

9. The 12th term of the AP $-5, -1, 3, \ldots$ is: 1

(a) 38 **(b)** 78 **(c)** 40 **(d)** 39

10. The sum of the first 50 natural numbers is: 1

(a) 2550 **(b)** 1276 **(c)** 1274 **(d)** 1275

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A spinner has 8 equal sectors numbered 1 to 8. Find the probability that it stops on a multiple of 3. 2

12. Why is tossing a coin considered a fair way to decide which team bats first? 2

13. Show that $2, 6, 18, 54, \ldots$ is a GP and write its nth term. 2

14. A taxi charges a fixed ₹200 and ₹40 per km. Write the fare after 1, 2, 3 km and find the fare for 10 km. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A card is drawn from a pack of 52. Find the probability of 3
(i) a black queen
(ii) the ace of spades
(iii) a number card from 2 to 10.

16. Find the first four terms of $s_1 = 3,\ s_n = s_{n-1}(s_{n-1} - 1)$. 3

17. A class of 40 has 24 girls. One student is chosen at random. Find the probability that the student is 3
(i) a girl
(ii) a boy.

OR

A die is rolled once. Find the probability that the number is both even and prime, and the probability that it is neither. 3

18. Show that $1, 3, 6, 10, \ldots$ is neither an AP nor a GP. 3

OR

The points (n, t_n) of the AP $2, 5, 8, \ldots$ are plotted. Show they lie on a straight line and find its slope. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A school raffle sells 1000 tickets and awards 5 prizes. Anil buys 10 tickets. 5
(i) Find the probability that one particular ticket wins a prize.
(ii) Find the probability that Anil's ticket set contains a winning ticket (assume at most one of his tickets can win).
(iii) Find the probability that a ticket does not win.
(iv) How many winning tickets are expected among 200 tickets bought?

OR

Case study: A cricket team played 20 matches: it won 12, drew 3 and lost 5. 5
(i) Find the experimental probability of winning, drawing and losing.
(ii) Check that they add to 1.
(iii) Estimate the number of wins in 30 matches.
(iv) Does this prove the team will win its next match?

20. **Case study:** A colony of 500 bacteria doubles every hour. 5
(i) Write the numbers at the end of hours 1 to 4.
(ii) Write the nth term.
(iii) How many bacteria after 6 hours?
(iv) After how many hours does the number first exceed 10,000?

OR

(i) Show that the sum of the first n odd numbers is n^2. 5
(ii) Find $1 + 3 + 5 + \cdots + 49$.
(iii) Express 144 as the sum of consecutive odd numbers starting at 1.

END OF QUESTION PAPER Total: 40 Marks

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